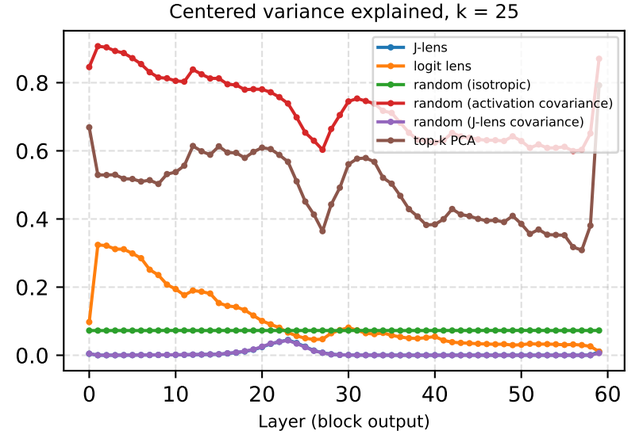
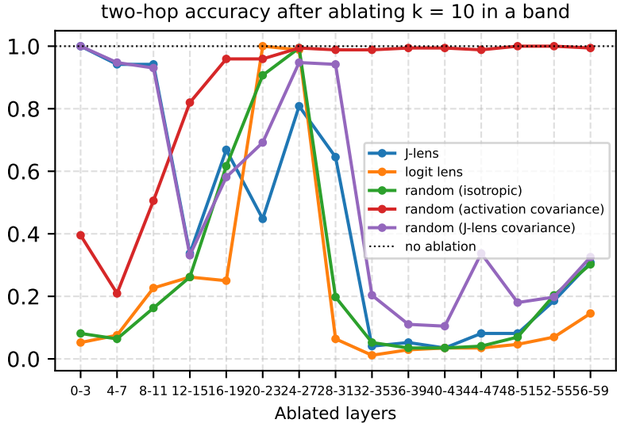
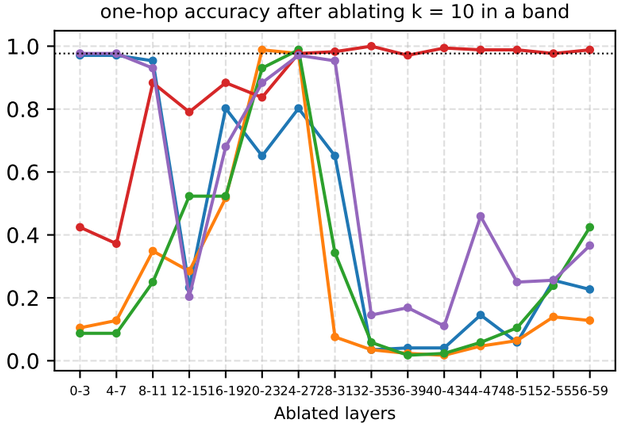
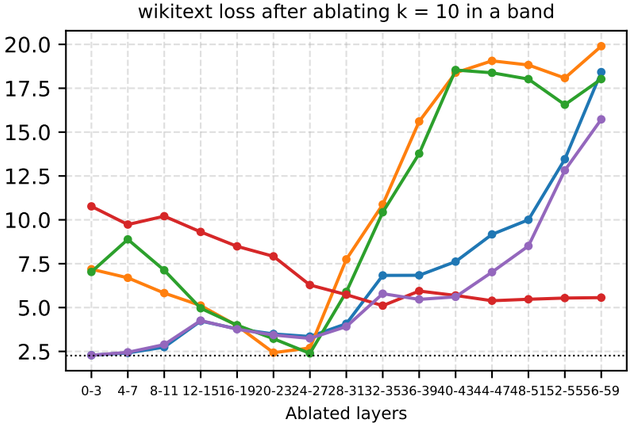
gemma-4-31b


Centered variance explained, k = 25
J-lens
logit lens
random (isotropic)
random (activation covariance)
random (J-lens covariance)
top-k PCA
0.0
0.2
0.4
0.6
0.8
0
10
20
30
40
50
60
Layer (block output)


two-hop accuracy after ablating k = 10 in a band
J-lens
logit lens
random (isotropic)
random (activation covariance)
random (J-lens covariance)
no ablation
0.0
0.2
0.4
0.6
0.8
1.0
0-3
4-7
8-11
12-15
16-19
20-23
24-27
28-31
32-35
36-39
40-43
44-47
48-51
52-55
56-59
Ablated layers


one-hop accuracy after ablating k = 10 in a band
0.0
0.2
0.4
0.6
0.8
1.0
0-3
4-7
8-11
12-15
16-19
20-23
24-27
28-31
32-35
36-39
40-43
44-47
48-51
52-55
56-59
Ablated layers


wikitext loss after ablating k = 10 in a band
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0-3
4-7
8-11
12-15
16-19
20-23
24-27
28-31
32-35
36-39
40-43
44-47
48-51
52-55
56-59
Ablated layers